AI & Automation Assignment (Black Friday Scenarios)

Theme 03: Insurance Perspective – Surge in Claims & Policy Coverage

Within each group, every participant must submit their own solution

Group 03	Sangeetha V
	Prakash E
	Nanjegowda YH
	Arjun Revuri
	S RAVI

Scenario: Black Friday returns result in **5% of all products sold (~2 million items)** being reported as damaged, lost, or stolen. Retailers rely on embedded insurance providers for **policy claims automation**. Manual claim adjudication timelines stretch to **10–15 days**, leading to customer dissatisfaction. AI-based claim validation and automation of settlement is needed to bring this down to **<48 hours**.

Problem Statement: How can AI-driven claims processing pipelines and automated coverage validation systems be architected to manage massive seasonal surges without increasing fraud exposure?

Assignment Instructions (Individual Work)

- **Objective:** Create an **Architecture Document** (using Arc42 template as reference) for your assigned scenario.

- **Steps:**
    1. Define the business challenge and context.
    2. Document functional & non-functional requirements (include NFRs, Metrics & KPIs).
    3. Propose architecture building blocks and interactions.
    4. Highlight where AI & automation augment the architecture (e.g., fraud detection, predictive analytics, infrastructure orchestration).
    5. Address risks, compliance, and trade-offs.
    6. Ensure traceability of design choices through **Architecture Decision Records (ADRs)**.

Submission: 4–6-page architecture document (Arc42-aligned) with diagrams, NFR table, and AI/automation integration highlights.

Best Practices & Frameworks to Discuss

- **Frameworks:**
    - **Arc42** for documentation structure.
    - **TOGAF ADM** for contextual alignment.
    - **ISO/IEC 25010** for NFR quality attributes.
    - **NIST AI RMF** for AI trust, fairness, and compliance.
    - **MLOps frameworks** for operationalizing AI in architecture.

- **Best Practices:**
    - Always document **AI/automation boundaries** (what's manual vs. automated).
    - Embed **metrics-driven design** (latency, accuracy, cost efficiency).
    - Ensure **explainability & traceability** for AI decisions (critical for BFSI & Insurance).
    - Design **resilience-first** (assume AI pipelines may fail, build graceful fallbacks).
    - Use **ADR rigor** to justify automation vs. manual interventions.